Perampalli Shravan Nayak

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LinkedIn | Website | GitHub | Hugging Face | Google Scholar

Research Summary

PhD student at Mila working on **agentic AI**: environments, data, and RL post-training for **computer-use and tool-use agents**. First author of open-source GUI agent benchmarks and datasets (UI-Vision, GroundCUA, VideoCUA) with **500k+ downloads**, and of EnterpriseOps-Gym, an environment suite for long-horizon tool use. Now focused on **proactive, personalized intelligence**: agents that act on a user's behalf and self-improve from their own experience.

Education

- **University of Montreal** — *Sep 2023 - Apr 2028 (Expected)*
  PhD in Computer Science — Montreal, Canada
  - Advisor: Prof. Aishwarya Agrawal | GPA: 4.3/4.3 | Fast-tracked from MSc (Sep 2025)

- **Indian Institute of Technology (BHU) Varanasi** — *Jun 2017 - May 2021*
  Bachelor of Technology in Electronics Engineering — Varanasi, India
  - GPA: 9.41/10 | Department Rank 1 (1st and 6th semesters) | Head, Club of Programmers

Experience

- **Microsoft Research, AI Frontiers** — *Jun 2026 - Aug 2026*
  Research Intern | *Host: Amanda Swearngin, Adam Fourney* — Redmond, WA, USA
  - Studied how agents can learn a user's routines, preferences, and open commitments by watching their computer screen
  - Built a scalable pipeline for evaluation environments that generates weeks-long user histories, compiles them into real web-app states, and renders screen recordings for agents to watch and act on
  - Developed novel proactive agent systems and agentic memory methods, and designed a benchmark to evaluate how well agents adapt to user preferences and offer help at the right moment [6]

- **ServiceNow Research** — *Aug 2024 - Apr 2026*
  Visiting Researcher | *Host: Sai Rajeswar, Spandana Gella* — Montreal, Canada
  - Created UI-Vision, a desktop GUI benchmark spanning 83 apps for evaluating computer-use agents [3]
  - Built GroundCUA (3.56M UI annotations) and VideoCUA (55 hours of humans performing computer-use tasks), and trained state-of-the-art grounding models with SFT and RL [2, 5]
  - Co-led EnterpriseOps-Gym, a benchmark of 1,150 enterprise tasks where frontier models solve under 50%, and built synthetic environments for training enterprise agents [1]

- **AMLRT, Mila – Quebec AI Institute** — *Mar 2024 - Jan 2025*
  Applied Science Intern | *Host: Hadrien Bertrand* — Montreal, Canada
  - Curated a pluralistic preference dataset of 37K pairwise comparisons across six community-defined criteria, and studied DPO for multi-preference steering of T2I models [9]

- **Microsoft** — *Jul 2021 - Aug 2023*
  Software Engineer II - PowerPoint Mobile Team — Hyderabad, India
  - Shipped features including background customization, text highlighting, and online video insertion
  - Reduced app hangs by 7.1% through memory optimizations and improved automation reliability by 16%

Publications

* Equal contribution

Computer-Use, Tool-Use & Multi-Agent Systems

[1] EnterpriseOps-Gym: Environments and Evaluations for Stateful Agentic Planning and Tool Use in Enterprise Settings [PDF] [Code]
Shravan Nayak*, Shiva Krishna Reddy Malay*, Jishnu Sethumadhavan Nair, ..., Sai Rajeswar [ICML 2026]

[2] Grounding Computer Use Agents on Human Demonstrations [PDF] [Code]
Shravan Nayak*, Aarash Feizi*, Xiangru Jian, Kevin Qinghong Lin, Kaixin Li, ..., Sai Rajeswar [ICLR 2026]

[3] UI-Vision: A Desktop-centric GUI Benchmark for Visual Perception and Interaction [PDF] [Code]
Shravan Nayak*, Xiangru Jian*, Kevin Qinghong Lin, Juan A Rodriguez, Montek Kalsi, ..., Sai Rajeswar [ICML 2025]

[4] Grammar Search for Multi-Agent Systems [PDF] [Code]
Mayank Singh, Vikas Yadav, Shiva Krishna Reddy Malay, Shravan Nayak, ..., Eduardo Blanco [ACL 2026]

[5] CUA-Suite: Massive Human-annotated Video Demonstrations for Computer-Use Agents [PDF] [Code]
Shravan Nayak*, Xiangru Jian*, Kevin Qinghong Lin, Aarash Feizi, Kaixin Li, ..., Sai Rajeswar [Under Review]

[6] From Experience to Initiative: Do Agents Understand a User by Watching Their Screen?
Shravan Nayak, Adam Fourney, Aishwarya Agrawal, Amanda Swearngin [In Preparation]

Reinforcement Learning, Post-Training & Alignment

[7] Discovering Failure Modes in Vision-Language Models using RL [PDF]
Kanishk Jain, Shravan Nayak*, Qian Yang*, Parisa Kordjamshidi, Nishanth Anand, Aishwarya Agrawal [Under Review]

[8] Value Drifts: Tracing Value Alignment During LLM Post-Training [PDF] [Code]
Mehar Bhatia, Shravan Nayak, Gaurav Kamath, Marius Mosbach, ..., Vered Shwartz, Siva Reddy [TACL 2026]

[9] LIVS: A Pluralistic Alignment Dataset for Inclusive Public Spaces [PDF] [Data]
Rashid Mushkani, Shravan Nayak, Hugo Berard, Allison Cohen, Shin Koseki, Hadrien Bertrand
Pluralistic Alignment Workshop, NeurIPS 2024 (Oral presentation) [ICML 2025]

Multimodal Evaluation, Interpretability & Robustness

[10] LatentLens: Revealing Highly Interpretable Visual Tokens in LLMs [PDF] [Code]
Benno Krojer, Shravan Nayak, Oscar Mañas, Vaibhav Adlakha, ..., Marius Mosbach [ICML 2026]

[11] CulturalFrames: Assessing Cultural Expectation Alignment in Text-to-Image Models and Evaluation Metrics [PDF] [Data]
Shravan Nayak, Mehar Bhatia, Xiaofeng Zhang, Verena Rieser, ..., Aishwarya Agrawal [EMNLP Findings 2025]

[12] Benchmarking Vision Language Models for Cultural Understanding [PDF] [Data]
Shravan Nayak, Kanishk Jain, Rabiul Awal, Siva Reddy, Sjoerd van Steenkiste, ..., Aishwarya Agrawal [EMNLP 2024, Oral]

[13] Improving Adversarial Robustness in Vision-Language Models with Architecture and Prompt Design [PDF]
Rishika Bhagwatkar, Shravan Nayak, Pouya Bashivan, Irina Rish
Trustworthy Multi-modal Foundation Models and AI Agents Workshop, ICML 2024 (Outstanding Paper) [EMNLP Findings 2024]

SKILLS

- **Research:** Computer-use & tool-use agents, agentic RL environments, LLM/VLM post-training, agent memory
- **Tools:** Python, PyTorch, Hugging Face (Transformers, TRL), verl, vLLM, multi-GPU training

AWARDS

- **NSERC Canada Graduate Research Scholarship – Doctoral (CGRS D)** ($40,000/year, 3 years) 2026–2029
- **FRQNT Doctoral Research Scholarship** ($25,000/year for 4 years, declined) 2026
- **Mila EDI Scholarship** ($8,000/year) 2024–2026
- **AI Research Scholarship** (Bourse en Intelligence Artificielle) 2025, 2026
- **Excellence Scholarships** (University of Montreal) 2024, 2025
- **DAAD WISE Scholarship** (research internship in Germany) 2020
- **Indian Academy of Sciences Summer Research Fellowship** 2019

SERVICE & TEACHING

- **Workshop Organizer:** MAPS @ CVPR 2026 [🔗], VLMs-4-All @ CVPR 2025 [🔗] 2025–2026
- **Reviewer:** NeurIPS, ICML, ICLR, CVPR, ACL, EMNLP 2024–2026
- **Teaching Assistant:** Representation Learning, University of Montreal 2024